

MCQ-Based Learning Platform

System Requirements Document

Version 1.0

Status	Draft for Review
Audience	Development Team, Product Owner
Scope	Sections 1 – 6

1. Overview

This document describes the functional requirements for a web-based Multiple Choice Question (MCQ) learning platform. The system supports multimedia-rich question delivery and enables teachers to manage content, enrol and register students, conduct chapter-based assessments, and evaluate student performance within a structured academic hierarchy. All users authenticate via a secure, role-based login system with email verification and password recovery.

2. User Roles

The platform defines three distinct user roles:

System Administrator (SA)	Platform super-user responsible for overall system governance. The SA manages teacher and student accounts, configures global platform settings, and has read access to all platform data for auditing and reporting purposes.
Teacher	Registered user who owns and manages academic content. A teacher creates subjects and chapters, authors MCQ questions, enrolls and registers students into chapters, conducts chapter-based exams, and reviews student performance. A teacher may manage multiple subjects, each containing multiple chapters.
Student	Registered user enrolled into one or more chapters by a teacher. Students log in to take assigned exams within their enrolled chapters, resume partial attempts, and view their results upon release.

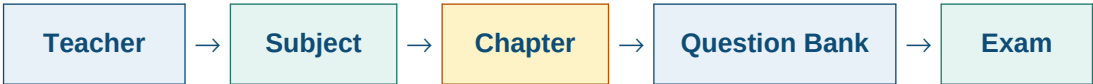
3. Authentication & Account Management

All three user types share a common authentication foundation:

- **Email verification** — accounts are not activated until the registered email address is confirmed
- **Password reset via email** — users may request a reset link at any time from the login screen
- Session management with configurable timeout
- The SA creates teacher accounts directly; students may be created by the SA **or by a teacher** during the enrolment/registration flow

4. Content Hierarchy

The platform follows a strict parent-child content structure:



Each teacher owns one or more subjects. Each subject contains one or more chapters. Each chapter maintains its own question bank and its own enrolled student list. Exams are chapter-scoped — only students enrolled in a chapter may sit that chapter's exams. Questions from the shared global bank may also be imported into a chapter's local bank.

5. Functional Requirements

5.1 System Administrator

- Create, update, deactivate, and delete teacher and student accounts
- Assign and modify user roles
- View platform-wide usage reports and audit logs
- Configure system-level settings (e.g. allowed media types, maximum file sizes, session timeout)
- Send platform-wide or role-targeted notifications

5.2 Teacher — Content Management

Create, edit, and delete subjects and chapters. Create MCQ questions within a chapter's question bank, each supporting one or more of the following media types attached to the question stem or to individual answer options:

- Plain text
- Image (uploaded file)
- Audio (uploaded file)
- Video (external link, e.g. YouTube or Vimeo)

Additional question settings:

- Define the correct answer and assign a positive mark value per question
- **Negative marking (optional)** — specify the mark to be deducted for an incorrect response, configurable per question
- Add an explanation or rationale for the correct answer, shown to students after result release (optional)

5.3 Teacher — Question Bank & Reuse

- Each chapter maintains its own **question bank** from which exams are assembled
- Teachers may access a **global shared question bank** containing questions across all their own subjects and chapters
- Teachers may **search, filter, and import** questions from the global bank into any chapter's local bank, avoiding duplication
- Imported questions are independent copies — edits to a copy do not affect the original
- Questions may be tagged by topic, difficulty level, or custom labels to aid search and reuse

5.4 Teacher — Student Enrolment & Registration

Teachers have full authority to bring students onto the platform and assign them to specific chapters without requiring SA intervention. The following workflows are supported:

Registering New Students

- A teacher may **create a new student account** directly from within any chapter's student management panel
- Required fields: full name, email address; the system auto-generates a temporary password and sends a welcome email with login credentials and an email-verification link
- The newly created student is **automatically enrolled** in the chapter from which they were registered
- The teacher may optionally enrol the same student in additional chapters at creation time or at any point afterwards
- Duplicate email addresses are rejected; the teacher is prompted to enrol the existing account instead

Enrolling Existing Students

- Search for any existing student on the platform by name or email address
- Enrol the found student in the current chapter with a single action
- Bulk enrolment via CSV upload — the file contains one student email per row; the system matches existing accounts and enrolls them, reporting any unmatched emails for follow-up
- A student may be enrolled in multiple chapters across different subjects simultaneously
- Teachers can only enrol students into chapters they own; cross-teacher enrolment is not permitted

Chapter Assignment & Management

- View the complete list of students currently enrolled in each chapter
- Move a student from one chapter to another within the same subject (the student's exam history in the original chapter is preserved but marked as transferred)
- Remove a student from a chapter enrolment — the student retains their account and any other chapter enrolments
- Re-enrol a previously removed student; their prior attempt history for that chapter is restored and remains visible
- Set an optional enrolment expiry date per student — access to the chapter and its exams is revoked automatically on expiry

Step	Action	Outcome
1	Teacher opens chapter → Student tab	Current enrolment list displayed
2a	Click 'Add Student' → search existing	Matching accounts listed; teacher selects and confirms
2b	Click 'Register New Student' → fill form	Account created; welcome email sent; student auto-enrolled
2c	Click 'Bulk Upload' → upload CSV	Matched accounts enrolled; unmatched emails flagged in report
3	Confirm enrolment	Student gains access to chapter and all published exams within it

All enrolment actions are logged with a timestamp and the acting teacher's identity for audit purposes.

5.5 Teacher — Chapter-Based Exam Management

Exams are scoped to a specific chapter. Only students currently enrolled in that chapter are eligible to sit its exams. This ensures content and assessment are always aligned to the correct cohort.

Creation & Configuration

Create an exam for a chapter by selecting questions from the chapter's question bank. The following settings are configurable per exam:

- **Chapter scope** — the exam is bound to one chapter; only enrolled students of that chapter may access it
- Total marks and passing score threshold
- Time limit (optional)
- Scheduled start and end date/time (availability window)
- **Attempt limit** — maximum attempts per student (once, fixed number, or unlimited)
- **Negative marking** — enable or disable at exam level; per-question deduction values apply when enabled
- Question and answer option shuffling (randomisation per attempt)
- Whether correct answers and explanations are shown to students after submission
- Whether results are immediately visible upon submission or held until teacher releases them

Lifecycle Management

- Save exams as draft, publish within a scheduled window, and archive completed exams
- Unpublish a published exam before it begins if changes are required
- Duplicate an existing exam to a different chapter as a starting template
- When a student is enrolled in a chapter that already has published exams, they gain access to all active exams immediately upon enrolment
- When a student is removed from a chapter, access to all unpublished exams in that chapter is revoked immediately

Monitoring & Results

- View all student submissions including attempt number, score, time taken, and submission timestamp
- View per-attempt answer breakdown for each student
- Manually release results to students in bulk or individually
- Export exam results as a CSV or spreadsheet file

5.6 Teacher — Analytics Dashboard

The analytics dashboard provides aggregated and per-student performance insights across subjects, chapters, and exams:

View	Description
Exam-level summary	Total attempts, average score, highest and lowest score, pass rate, average time taken
Question-level analysis	Per-question: % answered correctly, incorrectly, or skipped — identifies easy, hard, or ambiguous questions
Student progress	Per-student summary across enrolled chapters: exams taken, scores, pass/fail status, attempt history
Score distribution	Visual histogram of score ranges across all student attempts for a given exam
Trend analysis	Performance trends over multiple exam attempts or across chapters within a subject
Enrolment report	Chapter-wise headcount, enrolment date, expiry status, and participation rate (exams attempted vs available)

All dashboard views are filterable by subject, chapter, exam, date range, and individual student.

5.7 Student

Dashboard & Chapter Access

- View all chapters in which they are currently enrolled, along with the teacher's name and subject
- See the list of available (published) exams per chapter and their scheduled windows
- Chapters for which enrolment has expired are shown as read-only — past results remain visible but new exams cannot be accessed

Exam Access

- Begin an exam within its availability window, subject to the attempt limit configured by the teacher
- Only exams belonging to the student's enrolled chapters are visible and accessible

Partial Attempts & Resume

- If a student exits before submitting (browser close, session drop, or accidental navigation), in-progress answers and remaining time are automatically saved
- The student may **resume** the saved attempt from where they left off, within the same availability window and time limit
- A resumed attempt continues on the same attempt count — it is not treated as a new attempt
- If the time limit expires while the exam is paused, the attempt is automatically submitted with answers saved at that point

Results & Review

- View their result after the teacher releases it, including total score, marks breakdown per question, and pass/fail status
- View correct answers and explanations per question if the teacher has enabled answer review
- View full attempt history for an exam, including scores across multiple attempts if permitted

6. Notifications

The platform delivers in-app and email notifications to the relevant user upon the following events:

Event	Recipient
Account created by teacher	Student (welcome email with credentials)
Email verification required	Teacher / Student
Password reset requested	Teacher / Student
Enrolled into a chapter by teacher	Student
Assigned to a new chapter (transferred)	Student
Removed from a chapter enrolment	Student
Enrolment expiry reminder (e.g. 7 days before)	Student
Exam published and available	Student (chapter-enrolled only)

Exam window closing soon (e.g. 24 hours before)	Student
Exam result released	Student
Student submits an exam	Teacher (optional digest)
Platform-wide announcement	All users

Teachers may configure which exam-related notifications are active per chapter. The SA may configure system-level notification templates and delivery settings.